

Project Design Phase

Proposed Solution

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Project Name	Automated Network Request Management in ServiceNow

S.No.	Description Parameter	Proposed Solution Details
1.	Problem Statement (Problem to be solved)	• In traditional IT environments, network service requests (for new or existing connections, hardware links, and bandwidth allocations) are handled through manual emails, informal chats, or unstructured paper forms. • Requesters lack visibility into the status of their requests and face long waiting times. • IT and network teams spend excessive hours verifying missing details, manually entering data, and coordinating approvals, which leads to configuration backlogs, delayed fulfillment, and non-compliance with network security policies.
2.	Idea / Solution description	• Implemented a complete end-to-end automated Network Request Management system built on the ServiceNow Cloud Platform. • Self-Service Intake (/sp): Users submit a dedicated 'Network Request' catalog item under 'Network request Management'. It captures user details, contact numbers, Type of Connection (New/Existing), Total Amount, Mode of Payment (UPI/Card), Address, and file attachments, with dynamic Catalog UI Policies. • Data Persistence: Custom table 'Database Tables' (u_database_table) stores all records with unique Auto-Numbering. • 10-Step Flow Designer Logic: Automatically extracts variables, creates records, sends an instant acknowledgment email (Step 3), triggers 'Ask For Approval' (Step 4), and branches based on approval outcome—updating record state and dispatching

approval (Step 6/7) or rejection (Step 9/10) notifications.

3. Novelty / Uniqueness

- 100% Low-Code & Script-Free Automation: Entire workflow is executed using Flow Designer, avoiding complex custom scripts and enabling simple maintenance and visual troubleshooting via Flow Execution details.
- Context-Aware Dynamic Intake: Clean interface that displays only relevant inputs depending on whether the user selects a New or Existing connection and preferred payment method (UPI or Card).
- End-to-End Governance: Automatically creates an audit trail directly within platform approval records (sysapproval_approver) and keeps requesters informed through milestone email alerts at every stage.

4. Social Impact / Customer Satisfaction

- Eliminates User Anxiety: Requesters receive instant email receipts and milestone updates, removing the uncertainty of unacknowledged email requests.
- Reduced Turnaround Time: Automated routing cuts access delivery times from days to hours, ensuring employees, students, and engineers are not blocked from critical project work.
- Minimized Human Error: Mandatory form validation ensures complete technical details upfront, removing repetitive back-and-forth communication between engineers and users.

5. Business Model (Revenue Model)

- Internal IT Service Efficiency: Lowers operational overhead by reducing manual triage hours, allowing network engineers to focus on high-priority infrastructure and cybersecurity initiatives.
- Managed Service Provider (MSP) Model: Can be packaged and delivered as a modular Network-as-a-Service (NaaS) catalog offering for enterprise clients with per-request or tiered monthly subscription pricing.
- Direct Cost Recovery: Integrated capture of 'Total Amount' and 'Mode of Payment' (UPI/Card) enables transparent chargeback and billing across internal departments or external clients.

6. Scalability of the Solution

- Multi-Instance Cloud Scaling: Built natively on ServiceNow SaaS infrastructure, ensuring horizontal



scalability to process thousands of concurrent requests across multiple departments.

- **Extensible Architecture:** The custom table `u_database_table` and Flow Designer workflow can be extended to integrate with enterprise network orchestrators (e.g., Cisco DNA Center, Ansible) via ServiceNow Integration Hub for zero-touch port provisioning.
- **Robust Data Management:** Relational data indexing and default Access Control Lists (ACLs) guarantee high performance and strict data privacy as ticket volumes grow.